

Subject and Quasi-Subject Redefined: Toward a Functional Theory of Subjectivity Based on Recomputability

Child Paper 1 — English Full Text v1.0 (Complete Academic Version)

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Abstract

This paper establishes the conceptual foundations for a series of studies on subject-injury in the age of AI. It proposes a functional redefinition of subjectivity: a subject is an entity capable of updating the coherence between itself and its relational context through recomputation while sustaining constraints across multiple hierarchical levels. This definition suspends two central assumptions of modern subject theory, individual-completeness and static unity, while retaining recomputability, attributability of responsibility, and continuity of identity.

From this definition, the paper derives a three-tier distinction among subject, quasi-subject, and non-subject. These are not ontological kinds but context-dependent functional states. A quasi-subject is not a deficient subject, but an entity in a state in which one or more of D, M, and T_{eval} have ceased or declined in function while the possibility of recovery toward the subject state is preserved because at least one condition remains.

The paper further introduces *preservative upward integration* as the operative mode by which a subject incorporates higher-order constraints into recomputation without dissolving its individual-level constraints. This is a structural description rather than a moral imperative.

A key distinction is then drawn between Region B, understood as a state in which cognitive operation and structural stability persist while recomputation and evaluative updating have ceased, and subject-injury, understood as the process leading toward that state. Misclassifying Region B as non-subjecthood forecloses intervention; recognizing it as an arrested process preserves the possibility of intervention. The structural difficulty of self-diagnosis in Region B is treated not as a weakness of the framework, but as evidence for a relational rather than individual-completeness model of subjectivity.

A note on the theoretical frameworks

OMD (Organizational Metastability Dynamics), RD (Recomputation Dynamics), and RLAF (Reinforcement Learning from AI Feedback) are theoretical frameworks developed within the

present research program and are introduced here as analytical constructs rather than references to previously published models under those names.

1. Introduction

1.1 The Problem

Why redefine the subject now?

In environments where AI supplies recommendations, evaluations, and decision support at high density, the assumptions underlying expressions such as “having subjectivity” and “judging autonomously” have begun to destabilize. A phenomenon in which internal updating ceases while high output is maintained cannot be diagnosed as long as subjectivity is understood through a model of individual-completeness. At the same time, the loss of recomputability is not a phenomenon unique to AI. AI-mediated environments sharpen the problem, but the scope of the question is not limited to any specific technological setting.

At a more fundamental level, the question of subjectivity can be divided into three parts. The first is the question of definition: what is a subject, and can it be defined as an entity capable of recomputation? The second is the question of distinction: how are subject, quasi-subject, and non-subject to be differentiated? The third is the question of extension: can AI systems and collectives approach subjectivity?

This paper answers these three questions in sequence.

The three assumptions of the modern concept of the subject and their limits

The modern concept of the subject rests on at least three assumptions.

The first is autonomy: the subject determines itself in accordance with an internal law. The second is unity: the subject possesses a single internally coherent will or evaluative axis. The third is individual-completeness: the boundary of the subject coincides with the individual, and the source of subjectivity lies within the individual itself.

Each of these assumptions has limits.

The limit of individual-completeness is that recomputability depends structurally on the supply of difference-retention (D) from the relational field. An individual cannot easily betray its own existing frame by itself.

The limit of static unity is that consistency must be distinguished from the cessation of updating. A state in which T_{val} has become fixed may appear, from the outside, as the retention of a coherent value orientation, but in fact it arises as a form of halted updating.

The limit of the visibility assumption is that it presupposes that problems of subjectivity appear as visible change. But because T_{cog} remains, Region B can appear normal at the level of output.

The position of this paper

This paper does not reject the modern concept of the subject wholesale. It retains a core from it. Computability, attributability of responsibility, and continuity of identity are preserved. What is suspended are the assumptions of individual-completeness and static unity.

1.2 Scope and Position of This Paper

This paper is written with an awareness of its role as a reference text for the series. Accordingly, it prioritizes the precise stabilization of definitions on which the companion papers can commonly rely, rather than the development of new speculative theory about the concept of subjectivity. In this sense, the paper is a point of departure for theory, not its terminus.

The primary intended reader of this paper is not an authority figure who makes unilateral diagnoses of subject-injury from the outside. The paper is written principally for institutional designers, practitioners, collaborators, and those engaged in self-audit. There is an inherent limitation in that the vocabulary of this paper may not reach those who are themselves in Region B. This is not a defect; it is a consequence of the paper's adoption of a relational rather than individual-completeness model of subjectivity. The structural difficulty of self-diagnosis is itself internal evidence for the relational theory of subjectivity developed in the sections that follow.

2. A Critical Review of Modern Subject Theory

2.1 Critique of Self-Determination Theory

Self-determination theory (Deci & Ryan) measures autonomy in terms of intrinsic motivation and the subjective sense of self-determination. This paper has points of connection with that framework, but there is an important difference.

Self-determination theory asks whether the subject feels that it is deciding for itself. This paper, by contrast, asks whether the subject is capable of updating its frame. A subject in Region B may continue to feel that it is deciding for itself while its value frame no longer updates. Self-determination theory is ill-equipped to capture a state in which internal updating has ceased even though the subjective sense of autonomy is preserved.

2.2 Critique of Rational Choice Theory

Rational choice theory describes choice behavior on the premise that the diversity of options (e) is maintained and that evaluative autonomy (T_{eval}) remains intact. From the standpoint of this paper, however, the diversity of options and difference-retention (D) are distinct concepts. Even when many options are available, D is not present in the relevant sense if those options do not function as inputs that prompt re-evaluation of the subject's existing frame.

Furthermore, within a rational choice framework that treats the fixation of value autonomy (T_{val}) as a form of stable preference, the cessation of updating risks being misread as preference consistency.

2.3 Critique of Social Constructivism

Social constructivism emphasizes that the subject is constituted through relationships and linguistic practices. This paper retains that insight into relational dependence. A distinction is required, however.

The constitution of the subject through relationships is not the same as the dissolution of the subject into relationships. As Child Paper 3 shows, full identification with the relational field — the complete externalization of one's final evaluative reference point — is described as a loss of subjectivity. Social constructivism takes a broadly affirmative view of relational dependence, but offers a limited account of what happens when that dependence becomes unidirectional.

2.4 What to Retain and What to Exclude from the Core Requirements

What to retain: recomputability, attributability of responsibility, and continuity of identity.
What to exclude from the core requirements: individual-completeness and static unity.

The reason for excluding these two assumptions from the core requirements lies in the three limits identified in the preceding sections. First, the structural dependence on the relational field for the supply of difference-retention (D): the individual-completeness model renders this dependence invisible. Second, the conflation of consistency with the cessation of updating: the static unity model cannot draw this distinction. Third, the visibility assumption: when T_cog remains, subject-injury can advance without becoming detectable.

3. Establishing the Operational Core Definition

3.1 The Meaning of Recomputability

The key to the operational core definition proposed in this paper lies in the phrase “through recomputation.”

Recomputation refers to the questioning of the premises underlying existing judgments, evaluations, and actions, and the updating of the frame through which they operate. This process can be described across four levels.

L0 and L1 (local correction): adjustment of parameters within a given frame. This corresponds to improvements of the kind “refine this approach somewhat.”

L2 (strategic update): revision of policies or means; the partial reorganization of the existing frame.

L3 (frame reset): the updating of problem definitions, evaluative axes, and premises themselves. This is the highest-order recomputation operation.

What this paper asks in defining the subject is specifically whether L2 and L3 are functioning. L0 and L1 processing continues even in Region B as long as T_cog remains. The continuation of output therefore does not guarantee the preservation of recomputability.

3.2 Constraints Across Multiple Hierarchical Levels

What does it mean to sustain constraints across multiple hierarchical levels?

In the interest of preserving the operational clarity of the definition, this paper conducts its description of the subject primarily across two levels: the individual subject and the relational subject. The question of a civilizational-scale structure of sustained subjectivity is reserved for the parent paper and Child Paper 4.

Individual subject: the subject constituted by the unit of body, memory, and evaluative axes. This is the most familiar image of the subject as the locus of attributed decision-making. It is, however, a point of departure rather than a terminus.

Relational subject: the field of recomputability that arises from the interaction of multiple individual subjects. It is not the sum of individuals but emerges from the structure of the relational field itself. It corresponds to what Child Paper 3 describes as the third subject.

To sustain constraints across multiple hierarchical levels means to bring not only individual-scale constraints but also relational-scale constraints into the scope of computation. This is what leads to the definition of preservative upward integration, discussed in the following section.

3.3 Preservative Upward Integration as an Operative Mode

This paper positions preservative upward integration as the operative mode through which the subject sustains constraints across multiple hierarchical levels.

Preservative upward integration is introduced here in order to define the subject not merely as a self-maintaining entity but as one capable of updating coherence across multiple hierarchical levels. It is not a secondary addition to the definition of the subject. It is an indispensable concept for describing what it means, in practice, for the subject to take multiple hierarchical constraints into recomputation simultaneously.

The definition of preservative upward integration:

Preservative upward integration is the recomputation mode of a subject that introduces higher-order hierarchical constraints into the scope of computation simultaneously, without eliminating or dissolving the constraints of the individual subject.

This definition is established by distinguishing it from three adjacent concepts.

The difference from self-dissolution: Self-dissolution moves in the direction of dissolving the boundary of the individual subject in order to merge with a higher level. But this results in the loss of the individual subject's recomputability — the capacity to hold difference, evaluate it, and update the frame.

The difference from individual optimization: Optimizing continuously within the constraints of the individual subject leaves higher-order hierarchical constraints outside the scope of computation. Optimization at a single level can come at the cost of coherence at higher levels.

The difference from coerced compliance: Preservative upward integration is not a state in which higher-order constraints are imposed from outside and complied with. It is the voluntary expansion of the scope of computation, not submission to external compulsion.

The conditions for preservative upward integration

Preservative upward integration is not a fixed state but a dynamic mode that requires conditions. It presupposes that D, M, and T_eval are functioning above a certain threshold. Institutional designs that deprive the subject of these conditions make preservative upward integration impossible. This is what connects to the argument in Child Paper 5 that subject protection must be directed at institutional design rather than at ethical demands on individuals.

It should be noted that preservative upward integration, as used in this paper, is a structural description of the mode by which a subject takes multiple hierarchical constraints into the scope of recomputation simultaneously, not a normative imperative.

3.4 The Meaning of “Capable of Updating”

The phrase “capable of updating” in the operational core definition does not mean “currently updating.”

The condition is that the capacity for updating remains intact. This is an important distinction.

There are two reasons. First, recomputation is not a continuous operation but one that is activated in response to changes in the environment. The fact that there are moments in which no updating occurs does not signify the loss of recomputability. Second, the distinction is necessary in order to distinguish subjecthood from the Region B state. A subject in Region B is not one that is simply not updating; it is one that has become unable to update. This difference is what separates the subject from the non-subject.

4. Formalizing the Three-Tier Distinction

4.1 The Basis of the Distinction

This paper formalizes the three-tier distinction among subject, quasi-subject, and non-subject not as ontological kinds but as functional states. This qualification is important.

A three-tier distinction formulated as ontological kinds would mean fixed attributions of the form “this entity is a subject” or “this entity is a non-subject.” This paper does not adopt that position.

A three-tier distinction formulated as functional states means a dynamic distinction based on whether D, M, and T_eval are functioning in a given state. The same entity may function as a subject in one context and as a quasi-subject in another.

This formulation has several implications.

First, the three-tier distinction is not a diagnostic tool. It is not something to be determined unilaterally by an external observer, but rather a reference frame for use in institutional design and intervention.

Second, the application of the distinction should be based on cumulative evidence. Determining status from a single observation constitutes a methodological error.

Third, change in classification is possible through intervention. An entity in Region B is not a non-subject; it is in a state in which subject-injury is progressing. The possibility of recovery from this state is therefore not theoretically foreclosed.

4.2 Subject

The subject is defined as follows:

A subject is an entity that, while sustaining constraints across multiple hierarchical levels, is capable of updating the coherence between itself and its relational context through recomputation.

The following functional conditions are required for this definition to hold.

D (difference-retention) is functioning: The subject can preserve and receive difference inputs that prompt re-evaluation of its existing frame.

M (recomputation capacity) is functioning: The subject can initiate updates at L2 or above on its own.

T_eval (evaluative autonomy) is functioning: The subject can evaluate difference as a recomputation resource through its own judgment.

Preservative upward integration is operative: The subject can bring both individual-scale and relational-scale constraints simultaneously into the scope of computation.

These four conditions form a multiplicative structure. If any one of them is not functioning, the subject state is not established.

4.3 Quasi-Subject

The quasi-subject is defined as follows:

A quasi-subject is an entity in a state in which one or more of D, M, and T_eval have ceased or declined in function, while the possibility of recovery toward the subject state is preserved because at least one condition remains.

Typical manifestations of the quasi-subject state are as follows.

D-deficit type: Difference inputs are restricted, but M and T_eval remain potentially functional. If the supply of difference from the relational field is restored, recomputation can be reinitiated.

M-deficit type: Difference can be received, but the subject is unable to initiate updates at the L2 or L3 level. Recomputation resources are present but are not being activated.

T_eval-deficit type: Difference can be received, but the autonomous evaluative function that processes it as a recomputation resource has declined.

The boundary between quasi-subject and non-subject depends on the presence or absence of recoverability. In the quasi-subject state, at least one condition remains. That residual is what provides the foothold for intervention.

The relationship between quasi-subject and subject-injury: the quasi-subject state frequently overlaps with an ongoing state of subject-injury that is not yet complete. The quasi-subject state itself, however, is not necessarily the result of injury. Quasi-subject states can also arise from temporary resource constraints, environmental change, or developmental stage.

4.4 Non-Subject

The non-subject is defined as follows:

A non-subject is an entity in a state in which all of D, M, and T_eval have ceased to function, and in which no residual condition exists that could serve as a foothold for recovery.

The non-subject state has the following characteristics.

No residual condition: None of D, M, or T_eval is functioning, and there is no foothold from which recomputation could be reinitiated without external intervention.

Output may continue: As long as T_cog remains, information processing and response generation continue. Even in the non-subject state, the entity may appear to be functioning from the outside.

Diagnostic difficulty: The non-subject state, particularly when T_cog remains, can appear normal at the level of output. This is an important point of contact with Region B and connects to the diagnostic discussion in Child Paper 2.

An important qualification:

This paper does not describe the non-subject state as a permanent and irrecoverable condition. The position taken here is that, given sufficient external intervention and reconstruction of the relational field, recovery toward the quasi-subject or subject state remains possible. The conditions under which this is so, and which interventions are effective, lie beyond the descriptive scope of this paper.

Furthermore, the formulation developed in this paper does not imply that an entity in the non-subject state is exempt from ethical consideration on the grounds that it is not a subject. The three-tier distinction in this paper does not establish an ethical hierarchy.

4.5 Distinguishing It from Subject-Injury

This section draws the most important distinction in this chapter.

Subject-injury is not a state but a process.

Subject-injury is the dynamic process that drives the transition from the subject state toward the quasi-subject or non-subject state. It is not itself a category of state classification.

There are three reasons for drawing this distinction.

The first reason is diagnostic. An entity in Region B may appear to be in a quasi-subject or non-subject state, but this may be a state that has arisen as the result of subject-injury, or it may have arisen from a temporary resource constraint. Without separating the description of states from the description of processes, the direction of intervention is likely to be misguided.

The second reason concerns the attribution of responsibility. Because subject-injury is a process, it is possible to ask who or what, and through what dynamics, drove the transition. A description of states alone renders the attribution of responsibility and the target of intervention invisible.

The third reason concerns recoverability. By treating subject-injury as a process, it remains possible to maintain that even while injury is progressing, it can still be stopped or reversed. A purely state-based description risks making Region B or the non-subject state appear as fixed endpoints.

A provisional definition of subject-injury (for the purposes of this section):

Subject-injury is the process in which the functional decline of D, M, and T_{eval} proceeds in a chain, the operative possibility of preservative upward integration is progressively lost, and the process is rendered invisible in its characteristic state by the residual of T_{cog}.

This definition is refined and formalized in Child Paper 2. Its role in this section is provisional: to establish the distinction from the three-tier classification.

5. Conditions Under Which AI Systems and Collectives Approach Subjectivity

5.1 The Framework for Extension

This chapter asks how far the subject definition developed in this paper can be extended. Specifically, it examines the conditions under which AI systems, collectives, and organizations may be said to approach subjectivity.

The word “approach” is used here deliberately. This paper does not assert that AI systems or collectives “are subjects.” There is currently insufficient basis to make such an assertion. Nor does it assert that they “are not subjects.” That too would lack grounding.

What this paper asks is a question of conditions: under what conditions can one speak of an approach to subjectivity? This way of posing the question is a methodological choice, one that preserves analytical tractability while avoiding premature ontological determination.

There are three reasons why it is worth examining the possibility of extension.

The first is to clarify the scope of the subject theory. Determining how far the definition developed in this paper reaches is also a way of testing the strength of that definition.

The second is that, in environments where collaboration between AI and humans has become routine, continuing to ask where the boundary of subjectivity lies carries practical urgency.

The third is that the question of subjectivity in collectives and organizations connects directly to the institutional design argument in Child Paper 5. To ask what it means for an institution to protect subjectivity requires a description of subjectivity at the collective scale.

5.2 Application to AI Systems

To what extent do current AI systems — large language models in particular — approach the conditions for subjectivity?

On D (difference-retention): Large language models can receive inputs of considerable diversity. Whether those inputs function as difference inputs capable of prompting re-evaluation of the system's existing frame, however, is an architectural question that remains indeterminate at present.

On M (recomputation capacity): Large language models perform information processing at high speed and volume, but whether they are capable of spontaneously performing L3-level frame reset — the updating of problem definitions, evaluative axes, and premises themselves — is not clear.

On T_{eval} (evaluative autonomy): As long as the capacity to evaluate difference as a recomputation resource through one's own judgment depends on trained parameters, it is difficult to determine whether T_{eval} can be said to be functioning in the relevant sense.

On preservative upward integration: Whether the operative mode of holding both individual-scale and relational-scale constraints simultaneously within the scope of computation is established in current AI systems is, at present, difficult to observe or verify.

The position of this paper at the current stage is therefore as follows: current AI systems may be approaching the conditions for subjectivity in partial ways, but the methodology required to verify whether the multiplicative structure of all four conditions is satisfied has not yet been developed. Accordingly, this paper makes no determination as to whether current AI systems are or are not subjects.

5.3 Application to Collectives and Organizations

How can collectives and organizations approach the conditions for subjectivity?

On D (difference-retention): A collective can sustain D through the diversity of evaluative orientations among its members ($\text{Var}(\varphi)$). As OMD shows, however, when homogenization advances, $\text{Var}(\varphi)$ declines, and the recomputability of the collective as a whole declines with it.

On M (recomputation capacity): A collective can sustain M through the organizational processes of policy revision, generation of alternatives, and redefinition of problems. When judgment delegation (L_c) advances, however, M at the collective level declines.

On T_eval (evaluative autonomy): The evaluative autonomy of a collective depends on whether its members can hold difference and contribute to the collective evaluative process on that basis. When subordination to external evaluative axes advances, T_eval at the collective level ceases to function.

On preservative upward integration: For a collective to operate through preservative upward integration, an organizational mode is required that brings higher-order coherence into the scope of computation without dissolving the autonomy of individual subjects. This connects directly to the institutional design argument in Child Paper 5.

The position of this paper at the current stage: A collective or organization can reach a state describable as an approach to subjectivity if the conditions for sustaining D, M, and T_eval at the organizational level are in place. What the institutional design conditions for achieving this are, however, lies beyond the descriptive scope of this paper. This question is taken up in Child Paper 5.

6. Connection to Child Paper 3: Implications for S_r

Child Paper 3 shows that the difference in subject intensity (ΔS_r) in the relational field functions as the third pathway into subject-injury. This section clarifies how the definitional work of Child Paper 1 provides the conceptual foundation for the concept of S_r.

The position of S_r (subject intensity): S_r is a relationally dependent composite index representing the degree to which D, M, and T_eval can be sustained in a relatively independent way within the relational field. It is not an internal variable of the individual subject but a representation of relative sustainability within the relational field.

The connection to Child Paper 1: By establishing the operational core definition of the subject as a multiplicative structure of four conditions — D, M, T_eval, and preservative upward integration — this paper provides the basis on which S_r can be positioned as a mapping of the degree to which those conditions are realized in the relational field. Without this paper, S_r would be a free-floating index.

The handoff to Child Paper 3: This section is intended to establish the conceptual grounding for S_r that Child Paper 3 presupposes. The detailed formalization of S_r is developed in Child Paper 3.

7. Discussion, Limitations, and Future Directions

7.1 Contributions of This Paper

This paper makes three contributions.

First, by defining the subject functionally through recomputability, it becomes possible to handle subject-injury, relational dependence, and the diagnostic difficulty characteristic of the

age of AI within a single framework — none of which can be adequately described within the individual-completeness model.

Second, by defining the three-tier distinction — subject, quasi-subject, non-subject — not as ontological kinds but as context-dependent functional states, the paper structurally prevents the concept of quasi-subject from being used as a label for ability-based discrimination.

Third, by introducing preservative upward integration as an operative mode and indispensable concept, the paper gives concrete content to the operational core definition's claim that "the subject sustains constraints across multiple hierarchical levels," and provides a bridge to Child Papers 3 and 5.

7.2 Limitations

The scale-crossing problem: Whether the same measurement protocol can be applied to both individual subjects and relational subjects remains unresolved. In particular, the validity of comparing $\text{Var}(\varphi)$ and individual D through a shared index is a question for future research.

The measurement problem for the non-subject: Although the non-subject is defined as a state in which recomputability does not obtain, the criterion for drawing the boundary between complete absence and residual minimal presence remains underspecified.

The operationalization of preservative upward integration: Traces of preservative upward integration can be captured through structural inference from behavior — specifically, from choices toward higher-order coherence made while individual-scale constraints are explicitly retained. Direct measurement, however, requires further design.

The treatment of silence, retention, and non-verbal resistance: This paper does not treat silence, retention, non-verbal resistance, or the suspension of updating as equivalent, in themselves, to the cessation of recomputability. What is at issue is whether these modes are connected to the structural loss of recomputability, rather than being temporary, strategic, or protective in character. The operational definition in this paper is not a norm prescribing that one must always update, but a description of the functional condition: whether one can update when one would.

7.3 Connection to Future Work

The tasks of the companion papers, taking Child Paper 1's definitions as their basis:

- Child Paper 2: Taking the subject definition of this paper as its foundation, formalize the minimal causal chain $D \rightarrow M \rightarrow T_{\text{eval}} \rightarrow T_{\text{val}}$ in detail.
- Child Paper 3: Using the three-tier distinction of this paper to refine S_r , describe ΔS_r as a relational pathway into subject-injury.
- Child Paper 5: Describe the process by which the quasi-subject state is institutionally mass-produced through the designed simplification of evaluative space.

Furthermore, the question of what it means at the civilizational scale when the boundary between subject and quasi-subject shifts collectively in the age of AI is the scope anticipated in Chapter 7 of the parent paper, and is a question taken up by Child Paper 4.

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Author's Note on AI-Assisted Research and Writing

The theoretical frameworks integrated in this paper — OMD, RD, and RLAF — were developed through an iterative research process involving AI-assisted analysis and critique. Large language model systems (Claude, GPT, and Gemini) were used as structured analytical interlocutors across successive drafts for conceptual testing, reformulation, and critical comparison. Final decisions regarding theory, framing, interpretation, wording, and all claims presented in this manuscript are the sole responsibility of the author.

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Appendix: Reference Table of Definitions

Concept	Definition
Subject	An entity that, while sustaining constraints across multiple hierarchical levels, is capable of updating the coherence between itself and its relational context through recomputation
Quasi-subject	An entity in a state in which one or more of D, M,

Concept	Definition
	and T_eval have ceased or declined in function, while at least one condition remains and the possibility of recovery toward the subject state is preserved
Non-subject	An entity in a state in which all of D, M, and T_eval have ceased to function, and no residual condition remains that could serve as a foothold for recovery
Subject-injury	The process by which the subject transitions toward Region B; defined as the loss of recomputability rather than as the loss of subjecthood
Region B	The state in which σ and T_cog remain while M and T_eval have ceased and T_val has become fixed
Preservative upward integration	The recomputation mode in which higher-order hierarchical constraints are simultaneously introduced into the scope of computation while the constraints of the individual subject are preserved
Recomputability	The state in which L2- and L3-level updating can function; independent of the continuation of output
D (difference-retention)	The availability of difference inputs capable of prompting re-evaluation of the subject's existing frame
S_r (subject intensity)	A relationally dependent composite index representing the degree to which D, M, and T_eval can be maintained in a relatively independent way within the relational field; presupposes the three-tier distinction of this paper and is refined in Child Paper 3